provide air interface to the mobile station, that is, the user carrying the mobilephone fitted with a subscriber identity module (SIM). Other responsibilities handled by the base station include micro-mobility management such as handoff triggering, radio resource management and the like. A base station consists of several radio transmitters whose outputs are combined before being fed to an antenna and finally transmitted as EM waves.

## Measurement basics

The real source of EM radiation is the transmitting antenna installed at the telecom site, which determines EM field distribution in the vicinity of a transmitting station.

***Where to measure.*** Field properties of EM fields around the transmitting antenna are related to the field regions of the antenna. Therefore demarcation of an appropriate range of each field region is essential before starting an estimation of the field quantities.

Basically, the space surrounding an antenna is generally sub-divided into four regions: reactive near-field, reactive-radiating near-field, radiating (Fresnel) near-field and radiating far-field (Fig. 2). The reactive near-field region is immediately surrounding the antenna, and its radial distance from the centre of the antenna is $0.62(D^3/\lambda)^{1/2}$. Here, D is the largest dimension of the antenna and λ is the wavelength (to be valid, D must be large compared to wavelength). In this region, reactive field predominates.

At the boundary of the reactive near-field zone, a transition region exists wherein the radiating field, referred to as reactive-radiating near-field, begins to be important compared to the reactive component. Radial limit of the radiating near-field is $2D^2/\lambda$.

The region between reactive near-field and far-field region, wherein radiation field predominates, is referred to as Fresnel zone. This region exists only if maximum dimension D of the antenna is large as compared to wavelength λ. It is noteworthy to mention that radiation field predominates here, and propagation of EM waves is not a plane wave propagation. Electric (E) and magnetic (H) fields are locally normal, and ratio of $E/H \approx Z_0$ (intrinsic impedance of free space = 377Ω).

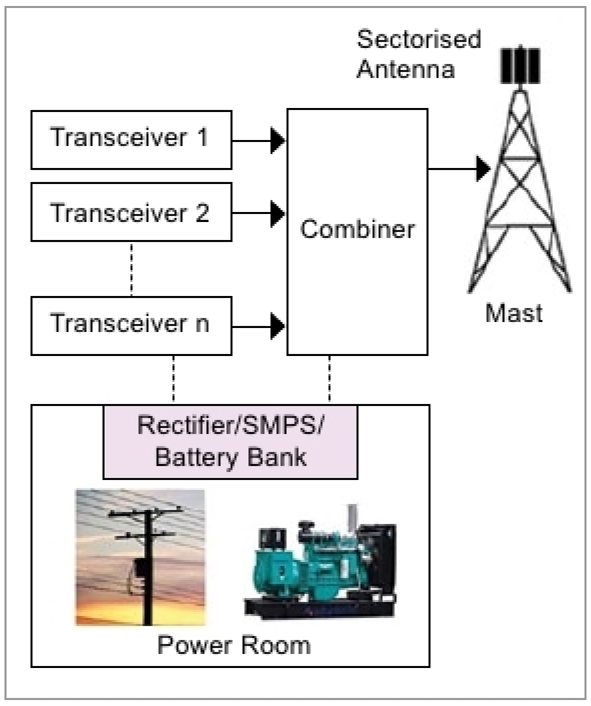


*Fig. 1: A typical mobilephone base station setup*

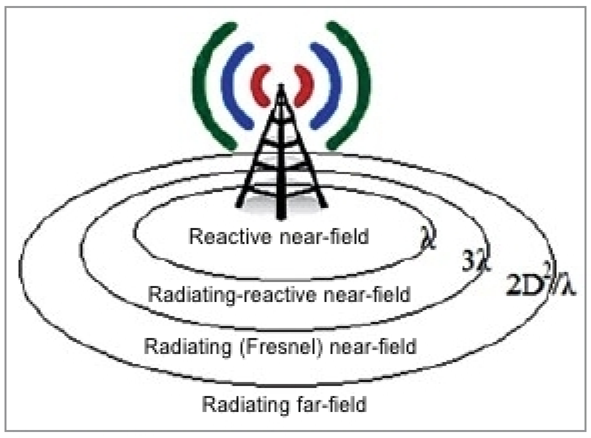


*Fig. 2: Field regions of an antenna*

Field region beyond radiating near-field (distance $\geq 2D^2/\lambda$ or larger value between 3λ and $2D^2/\lambda$) is radiating far-field, where field pattern is essentially independent of the distance from the antenna, and radiated power density is constant. In this region, electric (E) and magnetic (H) fields are perpendicular to each other, and the wave propagates as plane wave.

***What to measure.*** What is to be measured {E, H, power density (S) or specific absorption ratio (SAR)} depends on where (reactive or radiating field) the observer is, and on field impedance. Specific absorption ratio is the time derivative of the incremental energy absorbed by (dissipated in) an incremental mass contained in a volume element of a given mass density. It is expressed in units of watts per kilogram (W/kg).

In reactive near-field, it is required to measure both E and H components, or evaluate SAR. Determining SAR instead of field measurement is preferable at the positions located very close to the antenna. In reactive-radiating near-field, if no information on field impedance is available, it is suggested to measure both E and H fields. If information on field impedance is available, it is possible to measure only one field component. If $E/H > 120\pi$ (high-impedance EMF), measurement of E component is required. If $E/H < 120\pi$ (low-impedance EMF), measurement of H component will suffice.

Now, in radiating near-field, it is suggested to measure only E component with impedance taken equal to free space impedance ($Z_0$). In radiating far-field region, measurement of either E or H field component can be done to determine the equivalent power density. However, measurement of E field component is preferred. In far-field region, equivalent power density is obtained from the measured field by calculating, that is, by taking consideration of, Poynting vector.

## Some technical considerations

***Time averaging.*** Since mobile communication systems are highly time dependent, it is important to consider time averaging of the concerned fields. In this regard, ICNIRP has mentioned that RMS values of the fields are to be averaged over any six-minute period below 10GHz and over $68/f^{1.05}$-minute period for frequencies exceeding 10GHz (where f is frequency in GHz). In cases of expected time variability of the source, measurements are to be done for an extended period of time. In cases of channel variability, that is, load variability, measurements are to be conducted during time of peak usage.

***Multiple sources.*** In situations where there are multiple sources of radiations, the effect of multiple sources radiating EM waves at different frequencies is considered in a weighted sum. Each individual source is pro-rated according to the limit applicable to its frequency.

In most cases, a typical transmitting station contains many transmitting systems operating on many frequencies. In this scenario, all operating frequencies must be considered in a weighted sum, where each indi-